



Matheson Constructors Limited
245 Industrial Pkwy North
Aurora, Ontario L4G 4C4
P: (905) 669-7999
F: 905 669-0268

Project: 2025-048 Dockstader PRPS Reporting
Station
6 Dockstader Rd.
Brampton, Ontario L6R

Submittal #08 11 13-2.1 - Revised Door Frames, Doors, and Screens Shop Drawings

08 11 13 - STEEL DOORS AND FRAMES

Revision	1	Submittal Manager	Alvin Hernandez (Matheson Constructors Limited)
Status	Open	Date Created	Jul 29, 2026
Issue Date	Jul 29, 2026	Spec Section	08 11 13 - STEEL DOORS AND FRAMES
Responsible Contractor	Egress Systems of Canada Ltd.	Received From	Scott Minhinnick (Egress Systems of Canada Ltd.)
Received Date		Submit By	Nov 17, 2025
Final Due Date	Aug 12, 2026	Lead Time	84 day(s)
		Cost Code	
Location		Type	Shop drawing
Submittal Package			
Approvers	Alvin Hernandez (Matheson Constructors Limited), Ashish Kumar (Diamond Schmitt Architects)		
Ball in Court	Alvin Hernandez (Matheson Constructors Limited)		
Distribution	Ali Haidar (Matheson Constructors Limited), Alvin Hernandez (Matheson Constructors Limited), Arne Suraga (Diamond Schmitt Architects), Dhruv Patel (Matheson Constructors Limited), Hari Singh (Matheson Constructors Limited), Kathy Schaeffer (Matheson Constructors Limited), Marwan Diab (Matheson Constructors Limited), Marwan Sinno (Diamond Schmitt Architects), Tristan Crawford (Diamond Schmitt Architects)		
Description	Enclosed fore your review and acceptance is Submittal 08 11 13-02-R1-Revised Door Frames, Doors, and Screens Shop Drawings		
Schedule ID #			

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					
Scott Minhinnick		Jul 29, 2026	Jul 29, 2026	Submitted	Submittal 08 11 13-02-R1-Revised Door Frames, Doors, and Screens Shop Drawings ...pdf
Alvin Hernandez	Jul 29, 2026	Jul 29, 2026	Jul 29, 2026	Reviewed as noted	
Ashish Kumar	Jul 29, 2026	Aug 12, 2026	Aug 17, 2026	Reviewed as noted	304R1 08 11 13-2.1 Revised Door Frames, Doors, and Screens Shop Drawings_RN.pdf (Current) Transmittal_Submittal - 304R1 (Responded and Closed).pdf (Current)

Diamond Schmitt | 384 Adelaide Street West, Suite 100 Toronto ON M5V 1R7 Canada

PROJECT: Dockstader PRPS 221033 DATE SENT: 8/17/2026

SUBJECT: Revised Door Frames, Doors, and Screens Shop Drawings SUBMITTAL ID: 304R1

TYPE: Submittal TRANSMITTAL ID: 02609

PURPOSE: Reviewed as Noted VIA: Procore

SPEC SECTION: 08 11 13

FROM

NAME	COMPANY	EMAIL	PHONE
Ashish Kumar	Diamond Schmitt	akumar@dsai.ca	1-416-862-8800

TO

NAME	COMPANY	EMAIL	PHONE
Alvin Hernandez	Matheson Constructors	ashernandez@mathesonconstructors.com	905 669 7999
Ali Haidar	Matheson Constructors	ahaidar@mathesonconstructors.com	
Arne Suraga	Diamond Schmitt	ASuraga@dsai.ca	1 (416) 862-8800 x516
Ashish Kumar	Diamond Schmitt	akumar@dsai.ca	1-416-862-8800
Dhruv Patel	Matheson Constructors	dhpatel@mathesonconstructors.com	
Eric Charron	Diamond Schmitt	echarron@dsai.ca	1-416-862-8800
Fariz Dhalla		fariz.dhalla@entuitive.com	
Hari Singh	Matheson Constructors	hsingh@mathesonconstructors.com	
Haril Thakore		hthakore@cdml.ca	
Jamie Carmichael	CDML	jcarmichael@cdml.ca	1 (416) 583-2108
Kathy Schaeffer	Matheson Constructors	kschaeffer@mathesonconstructors.com	905 669 7999
Marwan Diab	Matheson Constructors	mdiab@mathesonconstructors.com	905 669 7999
Marwan Sinno	Diamond Schmitt	msinno@dsai.ca	1-416-862-8800
Tristan Crawford	Diamond Schmitt	TCrawford@dsai.ca	1-416-862-8800 ext. 641

Submittal Transmittal

DATE: 8/17/2026

ID: 02609

REMARKS:

CONTENTS

QUANTITY:	1	DATED:	7/29/2026	NUMBER:
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DESCRIPTION:

08 11 13-2-1 - Submittal Form.pdf

ACTION:

REMARKS:

QUANTITY:	1	DATED:	7/29/2026	NUMBER:
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DESCRIPTION:

Submittal 08 11 13-02-R1-Revised Door Frames, Doors, and Screens Shop Drawings ..pdf

ACTION:

REMARKS:

COPIES:

Ajit Stephen	(Region of Peel Deveolpment Services - Planning)
Majadeh Hamidsaeidi	(Region of Peel Deveolpment Services - Planning)
Jean-Louis Bramwell	(Total Opening Consultants)
Thennarasu Parimalam	(Total Opening Consultants)

Submittal #08 11 13-2.1 - Revised Door Frames, Doors, and Screens Shop Drawings

08 11 13 - STEEL DOORS AND FRAMES

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Received Date		Submit By	Nov 17, 2025
Final Due Date	Aug 12, 2026	Lead Time	84 day(s)
		Cost Code	
Location		Type	Shop Drawing
Submittal Package			
Approvers	Alvin Hernandez (Matheson Constructors Limited), Ashish Kumar (Diamond Schmitt Architects)		
Ball in Court	Ashish Kumar (Diamond Schmitt Architects)		
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Alvin Hernandez	Jul 29, 2026	Jul 29, 2026	Jul 29, 2026	Reviewed as noted	
Ashish Kumar	Jul 29, 2026	Aug 12, 2026		Pending	

245 Industrial Parkway North
Aurora, ON L4G 4C4
Phone: (905) 669-7999
Fax: 905-669-0268

Date: July 29, 2026 Job No.: 2025-048

To: Diamond Schmitt

Attn: Ashish Kumal

C.C.: Project File, A. Stephens, M. Sinno, K. Schaeffer, M. Diab, A. Haidar, H. Singh, D. Patel, A. Hernandez,

Re: Submittal 08 11 13-02-R1-Revised Door Frames, Doors, and Screens Shop Drawings

WE ARE SENDING YOU 1 electronic copy via email of the following items:

ZERO CARBON DOCUMENTS	SHOP DRAWINGS	PRODUCT INFORMATION	SAMPLES / MOCK UP	SUBSTITUTION REQUEST	WARRANTY		
	X						

NO.	DATE	REV #	DESCRIPTION
1	Jul. 29/26	1	Dwg. 1 of 11: General Notes, Abbreviations and Schedules
2	Jul. 29/26	1	Dwg. 2 to 5 of 11: Hollow Metal Frame and Door Schedule
3	Jul. 29/26	1	Dwg. 6 of 11: Frame Elevations
4	Jul. 29/26	1	Dwg. 7 of 11: Frame Elevations and Sections
5	Jul. 29/26	1	Dwg. 8 of 11: Screens
6	Jul. 29/26	1	Dwg. 9 of 11: Door Elevations
7	Jul. 29/26	1	Dwg. 10 of 11: Hollow Metal Sections
8	Jul. 29/26	1	Dwg. 11 of 11: Anchorage
9	Jul. 29/26	0	Guide Specification For Installation and Storage of Hollow Metal Frames.

THESE ARE TRANSMITTED as checked below:

☒ For approval

☐ For signature

☐ For your information

☐ As requested

☐ For review and comment

**diamond
schmitt**

In	2026-07-29	Out	2026-08-17
Project No.	221033	Specification Section No.	08 11 13
Reviewed	Reviewed as Noted MS	Revise and Resubmit	Not Reviewed

This review by Diamond and Schmitt Architects Incorporated is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of any engineering elements which form part of the contract documents prepared by others. This review shall not mean that Diamond and Schmitt Architects Incorporated approves the detailed design inherent in the submittal, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the submittal or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.



**TOTAL
OPENING
CONSULTANTS**

SHOP DRAWING REVIEW

This review is for the sole purpose of ascertaining conformance with the general design concept. The review shall not mean the detail design inherent in the shop drawings is approved. The review shall not relieve the Contractor of their responsibility for errors or omissions in the shop drawings or their responsibility for meeting requirements of the contract documents.

- ☐ REVIEWED
- ☒ REVIEWED AS NOTED
- ☐ REVISE AND RESUBMIT

BY: TP

DATE: 07-30-2026

REMARKS:

Alvin Hernandez

Signature

This transmission, including any accompanying attachments, contains information that may be privileged and/or exempt from disclosure under applicable law, and is intended only for the use of the recipient(s) named above. Any distribution, review, dissemination, or copying of the contents of this communication by anyone other than the intended recipient(s) is strictly prohibited. If you receive this communication in error, please notify the sender immediately and destroy the copy you have received. Thank you.

SECTION 08 11 13 SHOP DRAWING REV05

FOR

DOCKSTEADER PRPS

PEEL REGIONAL PARAMEDIC SERVICES
6 DOCKSTEADER ROAD, BRAMPTON, ON

This submittal is be
reviewed with architecture
and hardware consultants

ES25-015

VERSION DATE
29-Jun-2026 3:56 PM
SM25-023

GENERAL CONTRACTOR:

MATHESON CONSTRUCTORS
245 INDUSTRIAL PARKWAY NORTH
AURORA, ON. L4G 4C4
T: (905) 669-7999 F: (905) 669-0268

ARCHITECT:

DIAMOND SCHMITT ARCHITECTS INC
384 ADELAIDE ST. W. SUITE 100
TORONTO, ON. M5V 1R7
T: (416) 862-8800 F: (416) 862-5508

CONSULTANT

Scott Minhinnick
scott@egress-systems.com

PROJECT MANAGER

Mitch Minhinnick
mitch@egress-systems.com

EGRESS
SYSTEMS OF CANADA LTD

243 Saunders Road
Barrie, Ontario. Canada
L4N 9A3

T: (705) 733-3667

www.egress-systems.com

MC MATHESON CONSTRUCTORS	
Reviewed	()
Reviewed as Noted	(✓)
Revise and Resubmit	()
Not Reviewed	()
Date:	2026-07-29
Reviewed by:	
This review by MATHESON CONSTRUCTORS LIMITED is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of engineering elements which form part of the contract documents prepared by others. This review shall not mean that MATHESON CONSTRUCTORS LIMITED approves the detailed design inherent in the shop drawings, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the shop drawings or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.	



Missing:
A) D131.1

DSA: "Missing" doors noted
here are defined in the door
schedule on A051

B)D133.1A: Refer to PC-017

C)D302D,
D) D202C,

DSA: Door D302D does not exist in the project
Missing Screen SC08

Consultant: this submittal is to be
reviewed by Architect, Hardware.

Consultant: Refer to Submittal 08 11
13-01-R1-SR-009-Revised HM Doors
and Frames(RAN)

Subcontractor:
- Prep doors in accordance with the hardware schedule
- Prep Door frames to receive required junction boxes, conduits, etc.
- Coordinate glazing components with the Glazing Subcontractor.

DSA: Update subsequent
revisions to this submittal for
as-built and project records.

MATERIALS

Commercial grade steel to ASTM A653, CS, Type B, Coating Designation ZF75 (A40) Galvanneal
Galvanized G90 Material hot dipped ZF275 coating
Meeting all requirements of CSDMA

Minimum steel thicknesses in accordance with the Architects specification and Appendix 1 of the CSDMA

ZINC PRIMER APPLIED AT ALL DOOR, FRAMES AND SCREENS AT ALL WELDED AND GRINDED AREAS

GLASS AND GLAZING NOTES.

ALL DOORS, FRAMES, AND SCREENS ARE MANUFACTURED AS PER SPECIFICATION DIVISION 08 11 13
INTERIOR AND EXTERIOR SCREENS OR TRANSOM FRAMES TO BE GLAZED WITH ADJUSTABLE GLASS STOPS UP TO 25mm
PENDING FIRE RATINGS AND MAXIMUM GLASS SPACE AND THICKNESS REQUIREMENTS.
ALL SCREWS FOR THE GLAZING COME COMPLETE WITH HOLLOW METAL PRODUCTS.

PLEASE FOLLOW METAL DOOR SHOP DRAWINGS FOR GLASS SIZE THICKNESS AND LOCATION OF GLASS SIDE.
METAL DOOR LTD. IS NOT RESPONSIBLE FOR INCORRECT GLASS INSTALLATION, LOST SCREWS OR GLAZING STOPS.

ABBREVIATIONS

W/C	ZF120 (A40) GALVANNEAL
G90	ZF275 HOP DIPPED G90 GALVANIZED
HM	HOLLOW METAL
WD	WOOD
BL	BORROWED LITE
FR	FRAME
SC	SCREEN
PR	PAIR
L	LEFT
R	RIGHT
DA	DOUBLE ACTING
T	THERMALLY BROKEN
HT	HIGH TRAFFIC
EXT	EXTERIOR
INT	INTERIOR
HS	HOLLOW STEEL
HC	HONEY COMB

HOLLOW METAL TOLERANCES IS +/- 1.5 MM
ANY REQUIRED SITE MEASUREMENTS WILL BE
THE RESPONSIBILITY OF THE CONTRACTOR.

Dockstader PRPS

6 Dockstader Road, Brampton, ON
221033

File No. SPA-2024-0088

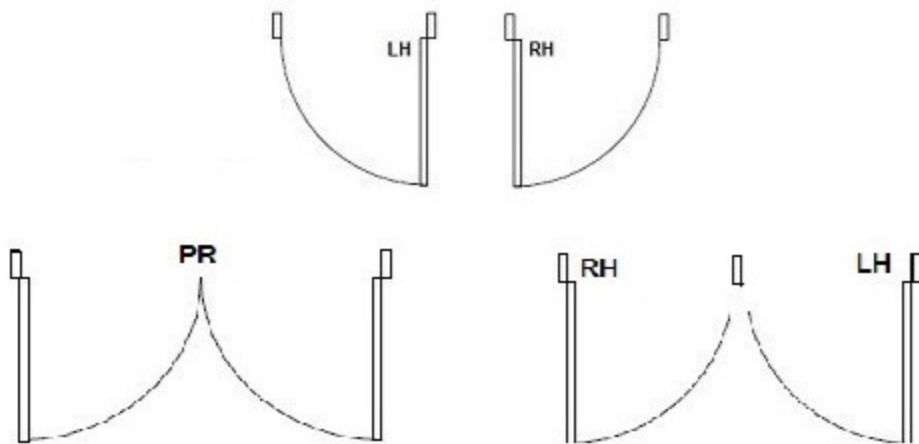
**diamond
schmitt**

MATHESON

EGRESS
SYSTEMS OF CANADA LTD

DATE: JUNE 22 2026	DIVISION 8 08 11 13	SUBMISSION 5
DRAWN BY DJL	S.O. # 45012 ES25-015-002	

SWINGS



Recognized By



Canadian Steel Door
Manufacturers Association
L'Association Canadienne
Des Fabricants de Portes D'Acier



Toronto
Construction
Association
Since 1887
Proudly Supporting Member



Door Security +
Safety Professionals

Canada



Intertek



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PLEASE VISIT WWW.CSDMA.ORG

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SYSTEMS OF CANADA LTD

ES25-015-002



METAL DOOR LTD



HOLLOW METAL FRAME AND DOOR SCHEDULE																														Job No: 45012			
	LOCATION				DOOR INFO												FRAME INFO												Revision Date: 29 Jun 2026 12:39				
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT THK	UND CUT	LABEL	DOOR LEAF DETAILS				PEMKO		ARCH	MD	PEMKO	THERMAL BREAK		WALL		REMARKS										
										SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R	SWING	TYPE	FINISH	Ga	ELEV	SECTIONS	RAIN DRIP	WD	HT	THCK	ANCH						
1	D1A-STA	SC	1		J	FG	HM	45	HT	19	3/4	L	1065	x	2440	FG	W/C	16		L	SC08	W/C	16	S1		15.E.G			1065 X 2440		T/A	SPRINKLER PROTECTED This screen is sprinkler protected, no label required	
2	D1A-STB	FR	1		C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16		L	F5	W/C	16	A		16			965 X 2440		S/F		
3	D1A-STD	FR	1		C	VL	HM	45	HT	19	3/4	R	965	x	2440	VL	W/C	16		R	F5	W/C	16	A		16			965 X 2440		S/F		
4	D1B-STD	FR	1	3/4 LABEL ADDED	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16		L	F2	W/C	16	A		16			965 X 2440		S/F		
5	D101C	FR	1	RUSH	A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		8			965 X 2440		T/A		
6	D107	FR	1	TOC: A?	A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F5	W/C	16	A		16			965 X 2440		S/F	60MFR is not an option, only 45MFR or 90MFR. In a 1hr partition, 45MFR is acceptable. 90MFR is for 2hr walls	
7	D108	FR	1		A		WD	45	INT	19		L	965	x	2440					L	F5	W/C	16	A		16			965 X 2440		S/F		
8	D109	FR	1		A		WD	45	INT	19		L	965	x	2440					L	F2	W/C	16	A		6			965 X 2440		C/S		
9	D116E	FR	1		C	VL	HM	45	HT	19		R	965	x	2440	VL	W/C	16		R	F2	W/C	16	A		10			965 X 2440		T/A		
10	D118C	FR	1		C	VL	HM	45	HT	19		L	965	x	2440	VL	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
11	D119A	FR	1		C		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S		
12	D119B	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S		
13	D121	FR	1		A	F	HM	45	HT	19	0HR	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
14	D122	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
15	D123	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		8			965 X 2440		T/A		
16	D125A	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		8			965 X 2440		T/A		
17	D125B	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
18	D126A	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
19	D126B	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		8			965 X 2440		T/A		
20	D127A	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
21	D130D	FR	1		D	G	HM	45	HT	19		L	965	x	2440	G	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A		
22	D131B	FR	1		C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16		L	F2	W/C	16	A		8			965 X 2440		T/A	60MFR is not an option, only 45MFR or 90MFR. In a 1hr partition, 45MFR is acceptable. 90MFR is for 2hr walls	
24	D2A-STB	FR	1		C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16		L	F5	W/C	16	A		16			965 X 2440		S/F		
25	D2A-STD	FR	1		C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16		L	F5	W/C	16	A		16			965 X 2440		S/F		
26	D2A-STE	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S		
27	D204	FR	1		A		WD	45	HT	19		L	965	x	2440					L	F5	W/C	16	A		16			965 X 2440		S/F		
28	D205	FR	1		A		WD	45	HT	19		L	965	x	2440					L	F5	W/C	16	A		16			965 X 2440		S/F		
29	D206	FR	1	3/4 LABEL ADDED	A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S		
30	D207A	FR	1	3/4 LABEL ADDED	L	F1	HM	45	HT	19	3/4	R	965	x	2440	F1	W/C	16		R	F2	W/C	16	A		12			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR 665 X 400 CHECK HARDWARE FOR CUTOUT SIZES	
31	D207B	FR	1	3/4 LABEL ADDED	L	F1	HM	45	HT	19	3/4	L	965	x	2440	F1	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR 665 X 400 CHECK HARDWARE FOR CUTOUT SIZES	
32	D208	FR	1		A		WD	45	HT	19		L	965	x	2440					L	F2	W/C	16	A		12			965 X 2440		C/S		
33	D209	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		12			965 X 2440		C/S		
34	D213B	FR	1	212 TO 213	A		WD	45	HT	19		R	1220	x	2440					R	F2	W/C	16	A2		13F			1220 X 2440		C/S	CASE OPEN SLIDER FRAME	
35	D215	FR	1		A		WD	45	HT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S		
36	D215B	FR	1	WASH TO 215	A		WD	45	INT	19		R	915	x	2440					R	F2	W/C	16	A2		12F			915 X 2440		C/S	CASE OPEN SLIDER FRAME	
37	D216	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S		

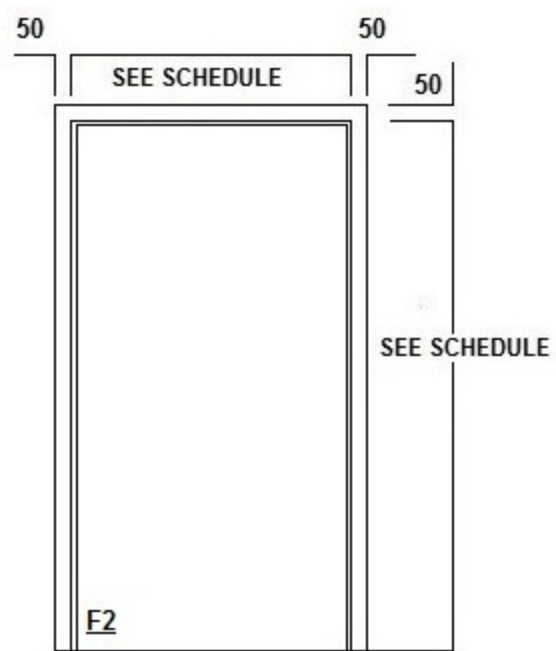
HOLLOW METAL FRAME AND DOOR SCHEDULE																											Job No: 45012				
	LOCATION				DOOR INFO										FRAME INFO												Revision Date: 29 Jun 2026 12:39				
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS				PEMKO		ARCH			MD	PEMKO	THERMAL BREAK		WALL		REMARKS					
											SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R	SWING	TYPE	FINISH	Ga	ELEV	SECTIONS	RAIN DRIP	WD		HT	THCK	ANCH		
38	D217	FR	1	TOC: A?	A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		11		965 X 2440		C/S	LOUVRE CUTOUT IN DOOR 665 X 400 CHECK HARDWARE FOR CUTOUT SIZES
39	D219	FR	1		L	F1	HM	45	HT	19	0HR	L	965	x	2440	F1	W/C	16		L	F2	W/C	16	A		12		965 X 2440		C/S	
40	D220	FR	1	DELETED	C		WD	45	INT	19		L	965	x	2440					L	F4	W/C	16	A		DELETED		965 X 2440		*	
41	D221	FR	1		D		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13		965 X 2440		C/S	
42	D222	FR	1		A		WD	45	INT	19		L	965	x	2440					L	F2	W/C	16	A		13		965 X 2440		C/S	
43	D223	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		13		1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16													
44	D224	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13		965 X 2440		C/S	
45	D225	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		6		1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16													
46	D227	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13		965 X 2440		C/S	
47	D229A	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		13		1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16													
48	D229C	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13		965 X 2440		C/S	
49	D230	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13		965 X 2440		C/S	
50	D231	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13		965 X 2440		C/S	
51	D232	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13		965 X 2440		C/S	
52	D233	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13		965 X 2440		C/S	
53	D235	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		12		965 X 2440		C/S	
54	D236	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12		965 X 2440		C/S	
55	D237	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		12		965 X 2440		C/S	
56	D238	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12		965 X 2440		C/S	
57	D239	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		12		965 X 2440		C/S	
58	D3A-STD	FR	1		A	F	HM	45	INT	19	3/4	R	965	x	2135	F	W/C	18		R	F5	W/C	16	A		16		965 X 2135		S/F	
59	D3B-STD	FR	1	3/4 LABEL ADDED	A	F	HM	45	INT	19	3/4	R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		13		965 X 2135		T/A	
60	D302C	FR	1		A	F	HM	45	HT	19	3/4	L	1067	x	2440	F	W/C	16		PR	F2	W/C	16	B		13		2134 X 2440		T/A	
				TOC: D302D door missing DSA: No D302D in project								R	1067	x	2440	F	W/C	16													
61	D303	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		12		965 X 2135		C/S	
62	D304	FR	1	DELETED	A	F	HM	45	INT	19		L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		DELETED		965 X 2135		C/S	
63	D305	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		12		965 X 2135		C/S	
64	D306	FR	1		A	F	HM	45	INT	19	3/4	L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		12		965 X 2135		C/S	
65	D307	FR	1		A	F	HM	45	INT	19	3/4	L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		12		965 X 2135		C/S	
66	D1A-STE	FR	1		A	F	HM	45	EXT	19		L	965	x	2440	F	G90	16		L	F1	G90	16	A		T8	☑	965 X 2440		S/F	
67	D1C-STD	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☑	965 X 2440		S/F	
68	D101B	FR	1	RUSH	A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☑	965 X 2440		S/F	
69	D102F	FR	1	RUSH	A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☑	965 X 2440		S/F	
70	D116A	FR	1	RUSH	C	VL	HM	45	EXT	19		R	965	x	2440	VL	G90	16		R	F1	G90	16	A		T14	☑	965 X 2440		S/F	

HOLLOW METAL FRAME AND DOOR SCHEDULE																											Job No: 45012						
	LOCATION				DOOR INFO								FRAME INFO														Revision Date: 29 Jun 2026 12:39						
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS				PEMKO		ARCH SWING	TYPE	FINISH	Ga	MD ELEV	PEMKO SECTIONS	THERMAL BREAK		WALL								
											SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R						RAIN DRIP	WD	HT	THCK	ANCH	REMARKS					
71	D130A	FR	1		D	G	HM	45	EXT	19		L	965	x	2440	G	G90	16			L	F1	G90	16	A		T7	☒	☐	965 X 2440		S/F	
72	D131A	FR	1		C	VL	HM	45	EXT	19		L	965	x	2440	VL	G90	16			L	F1	G90	16	A		T8	☒	☐	965 X 2440		S/F	
73	D138	FR	1	3/4 LABEL REMOVED	A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16			R	F1	G90	16	A		15	☐	☐	965 X 2440		S/F	
74	D229B	FR	1		A	F	HM	45	EXT	19		L	965	x	2135	F	G90	16			L	F1	G90	16	A		T14	☒	☐	965 X 2135		S/F	
75	D3A-STB	FR	1	3/4 LABEL REMOVED	A	F	HM	45	EXT	19		L	965	x	2135	F	G90	16			L	F1	G90	16	A		T14	☒	☐	965 X 2135		S/F	DOOR HEIGHT CHANGED TO 2135
76	D302A	FR	1		A	F	HM	45	EXT	19		R	1065	x	2050	F	G90	16			R	F1	G90	16	A		T14	☒	☐	1065X2050		S/F	DOOR HEIGHT CHANGED TO 2050
77	D302B	FR	1		A	F	HM	45	EXT	19		R	1065	x	2135	F	G90	16			R	F1	G90	16	A		T7	☒	☐	1065X2135		S/F	
78	RH-01	FR	1	DELETED By others	A	F	HM	45	EXT	19		L	915	x	915	F	G90	16			L	A1	G90	16	A1		DELETED	☒	☐	915 X 915		S/F	ROOF HATCH 4 SIDED FRAME By Roofer
81	SC09-17	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
82	SC09-16	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
83	SC09-15	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
84	SC09-14	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
85	SC09-13	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
86	SC09-12	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
87	SC09-11	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
88	SC09-1	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
89	SC09-9	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
90	SC09-18	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
91	SC09-7	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
92	SC09-6	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
93	SC09-5	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
94	SC09-4	SC	1						INT												SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
95	SC09-3	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
96	SC09-2	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
97	SC09-19	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
98	SC09-8	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
99	SC09-10	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
100	SC09-21	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
101	SC09-20	SC	1						INT												SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED	
102	SC10-13	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
103	SC10-12	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
104	SC10-11	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
105	SC10-10	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
106	SC10-9	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
107	SC10-1	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
108	SC10-8	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
109	SC10-14	SC	1						INT												SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED	
Print Date: Jun 29 2026																											METAL DOOR LTD. P.O. Box 638, Gormley, Ontario. L0H 1G0 www.metaldoorltd.com		Page 4 of ____				

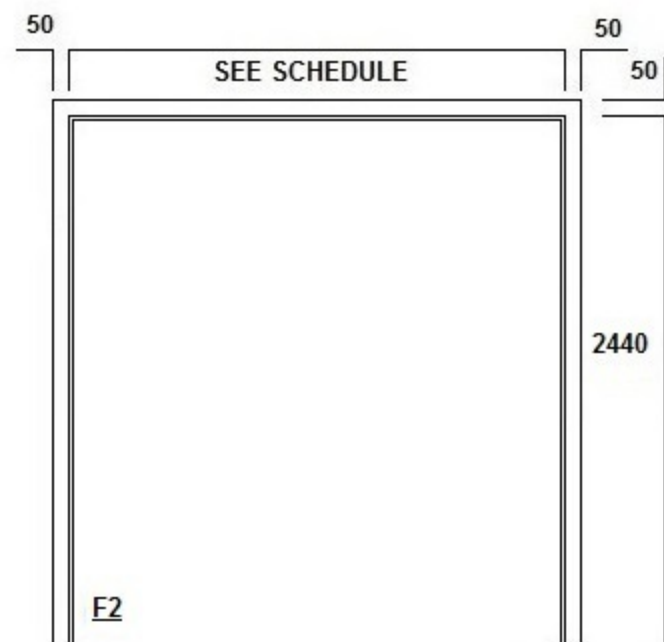
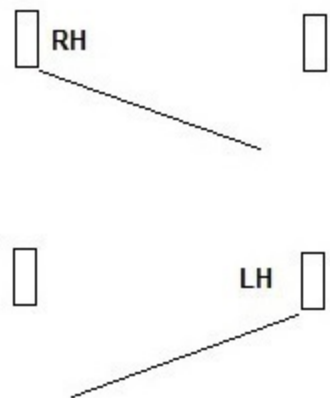
HOLLOW METAL FRAME AND DOOR SCHEDULE																									Job No: 45012								
	LOCATION				DOOR INFO								FRAME INFO												Revision Date: 29 Jun 2026 12:39								
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT THK	UND CUT	LABEL	DOOR LEAF DETAILS				PEMKO		ARCH				MD	PEMKO	THERMAL BREAK		WALL		REMARKS							
										SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R	SWING	TYPE	FINISH	Ga	ELEV	SECTIONS	RAIN DRIP	WD	HT			THCK	ANCH				
110	SC10-7	SC	1						INT										SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
111	SC10-6	SC	1						INT										SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
112	SC10-5	SC	1						INT										SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
113	SC10-4	SC	1						INT										SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
114	SC10-3	SC	1						INT										SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
115	SC10-2	SC	1						INT										SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
116	SC14	SC	1	Glass Type is GL-11					INT										SC14	W/C	16	S4		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
117	SC15	SC	1	HOLD PENDING CHANGE					INT										SC15	W/C	16	S5		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED			
118	D202C	FR	1	EXTRA	J	FG	HM	45	HT	19	3/4	L	1067	x	2440	FG	W/C	16			PR	B	W/C	16	B		7			2134 X 2440	156	C/S	

Proposed change is to be issued for SC 15

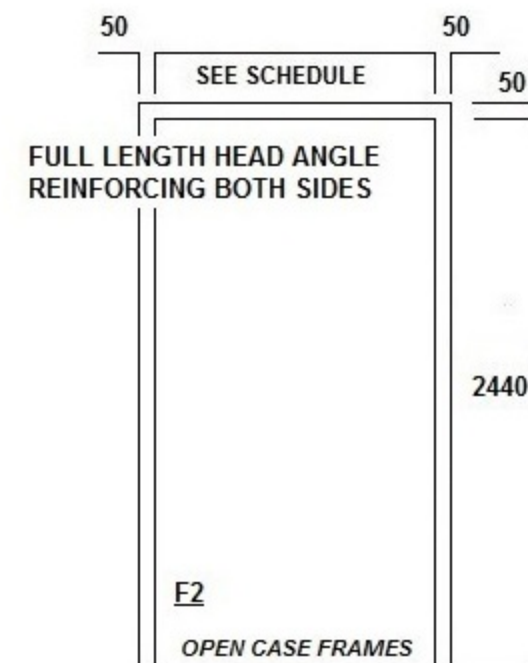
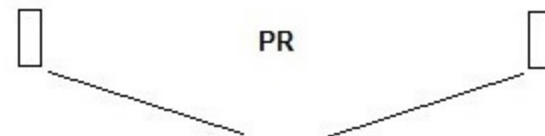
Glass Type is GL-11



ELEV A
68 REQ'D

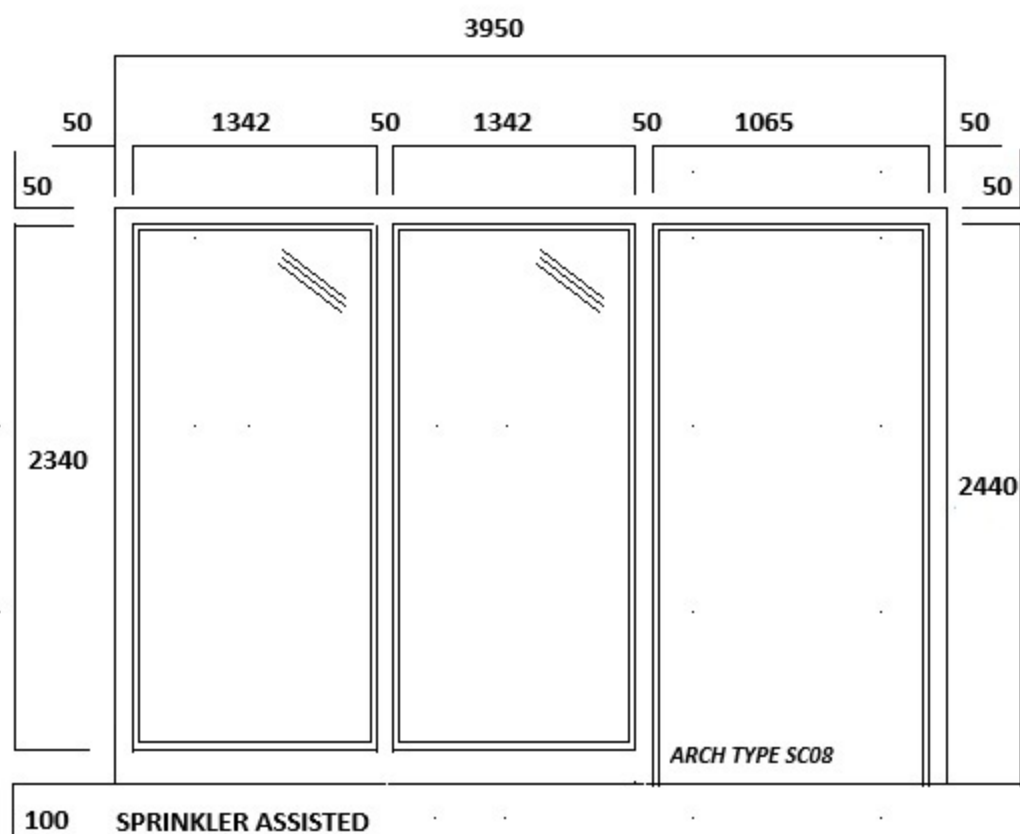


ELEV B
5 REQ'D

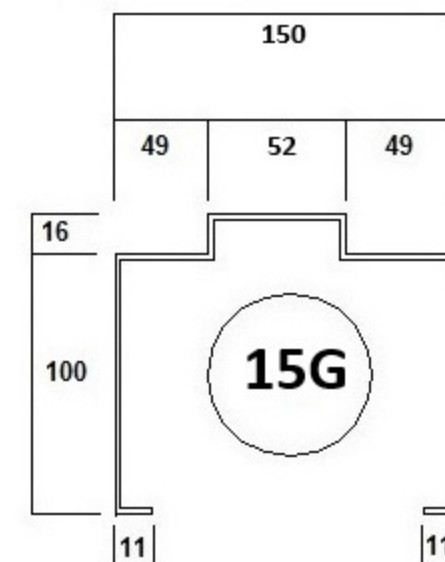
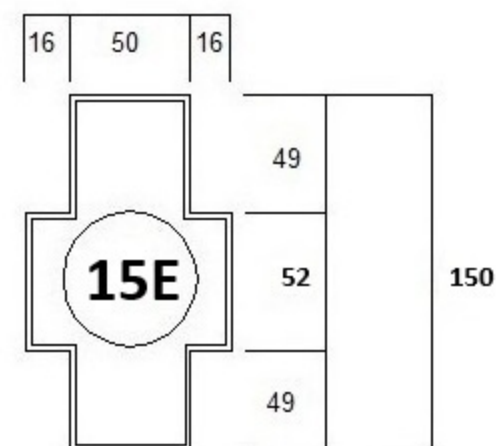
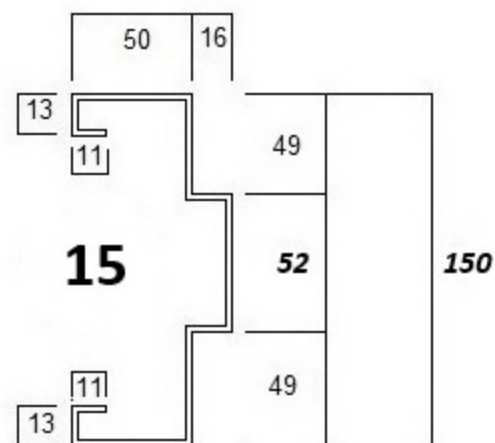
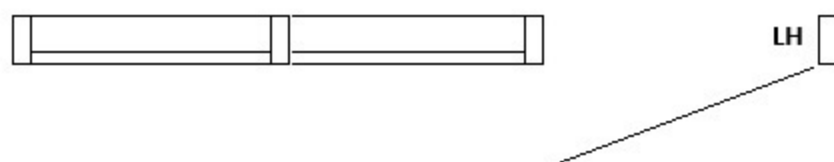


ELEV A2
2 REQ'D

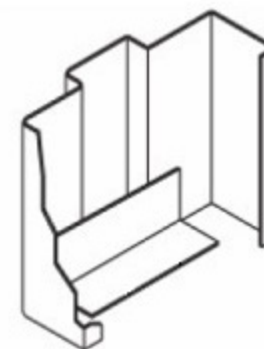




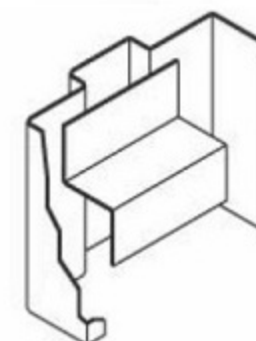
ELEV S1
1 REQ'D
D1A-STA



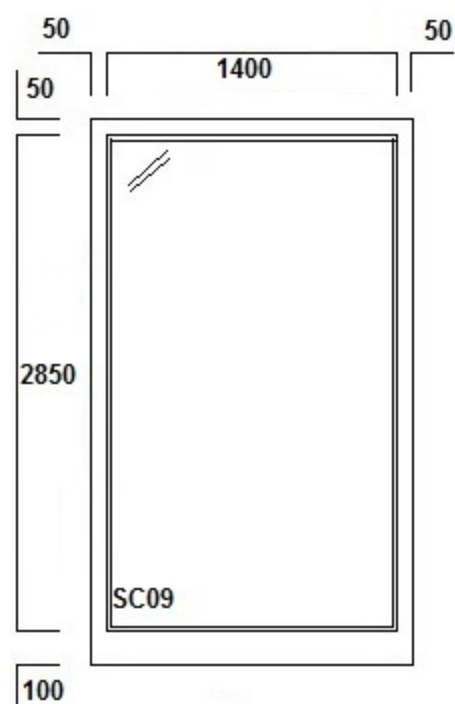
ADJUSTABLE MULLION BASE ANCHOR
SHIPPED IN MULLION OR LOOSE



WELDED FLOOR ANCHOR (T/A)

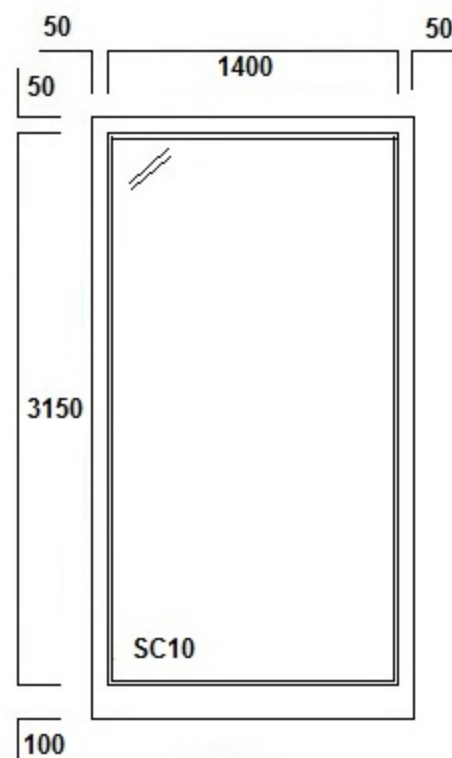


DRYWALL CHANNEL STUD ANCHOR (C/S)
WITH WELDED (T/A)



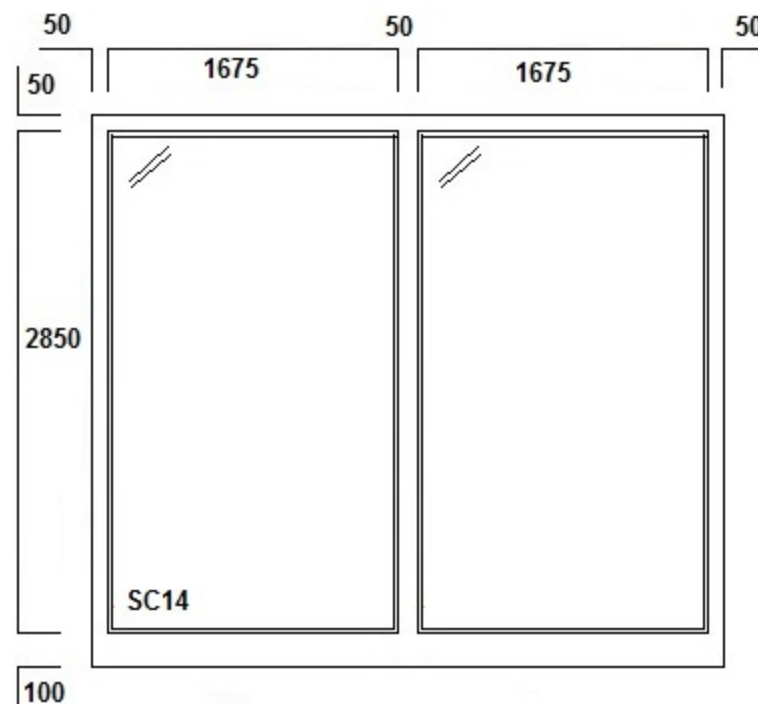
ELEV S2
21 REQ'D

SPRINKLER PROTECTED



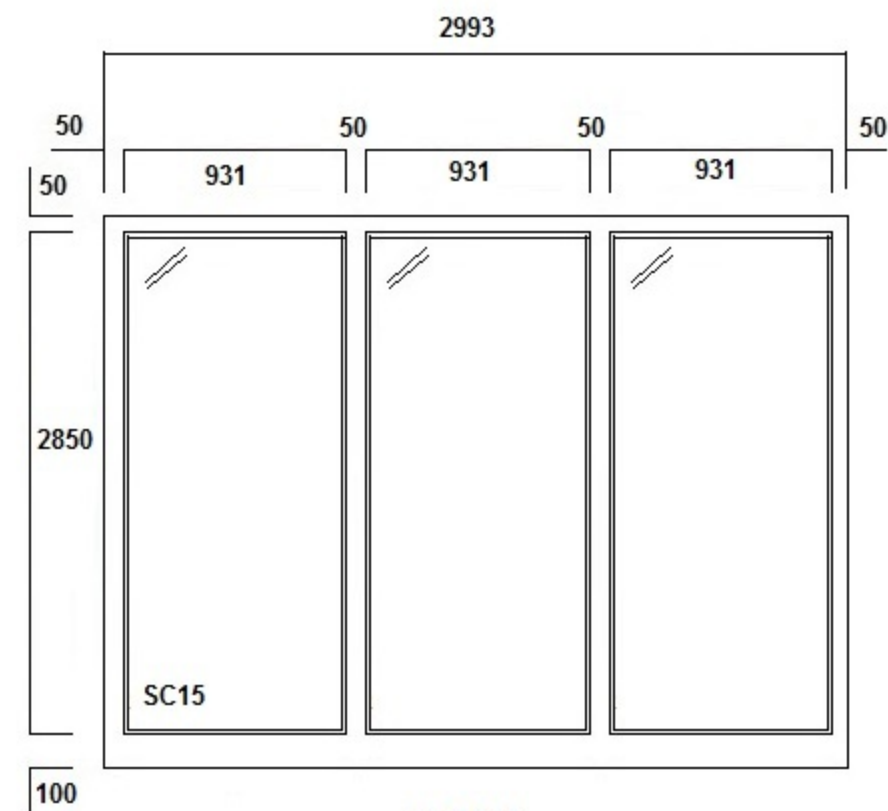
ELEV S3
14 REQ'D

SPRINKLER PROTECTED



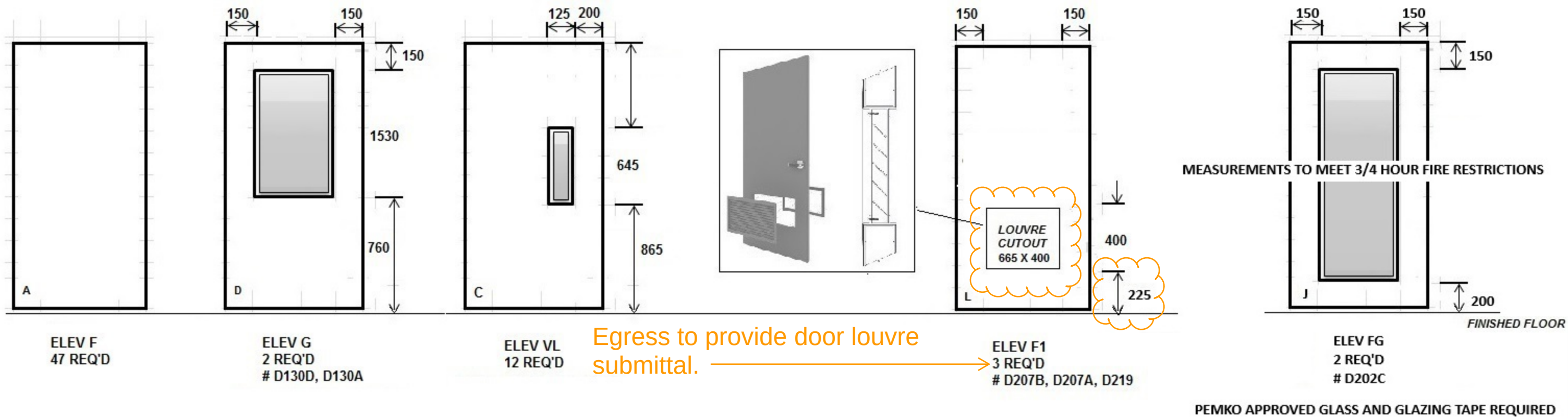
ELEV S4
1 REQ'D

SPRINKLER PROTECTED



ELEV S5
1 REQ'D

SPRINKLER PROTECTED



Egress to provide door louvre submittal.

INTERIOR DOORS
18 GA A40 GALVANNEAL
WELDED AND FILLED INTERLOCKING SEAMS GROUND SMOOTH
HONEYCOMB BOARD INSULATION

INTERIOR GLASS STOPS
UP TO 6MM GLASS THICKNESS

TACK AND FILL

FIRE RATED
UL LISTED

EXTERIOR DOORS
16 GA ZF275 G90 GALVANIZED
WELDED LAP SEAM
STEEL STIFFENED
URETHANE FOAM INSULATION

EXTERIOR GLASS STOPS
REMOVABLE SCREW IN GLASS STOPS
UP TO 25MM NON RATED GLASS THICKNESS

IT IS NOT RECOMMENDED TO PAINT EXTERIOR DOORS A DARK COLOR

FULLY WELDED

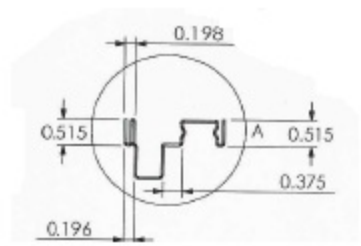
FLUSH STEEL TOP CAPS

HIGH TRAFFIC DOORS
16 GA A40 GALVANNEAL
FULLY WELDED LAP SEAM
STEEL STIFFENED
POLYISOCYANURATE CORE

INTERIOR GLASS STOPS
UP TO 6MM GLASS THICKNESS

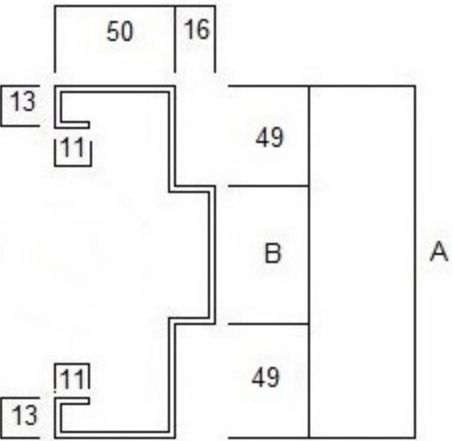
FULLY WELDED

FIRE RATED
UL LISTED



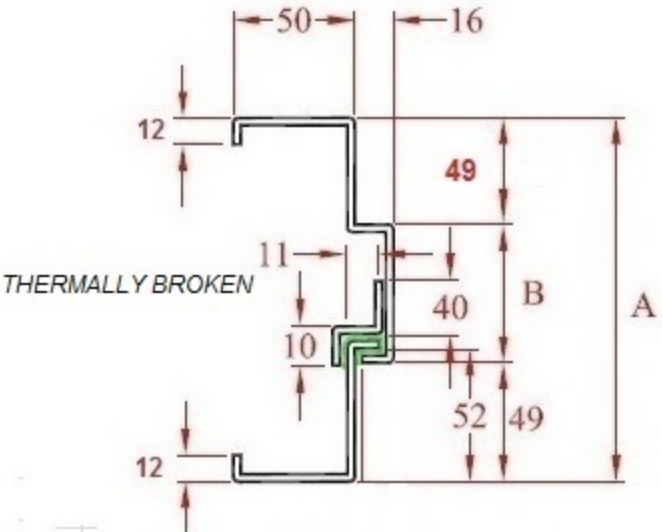
INTERIOR DOOR GLAZING DETAIL

HOLLOW METAL SECTIONS



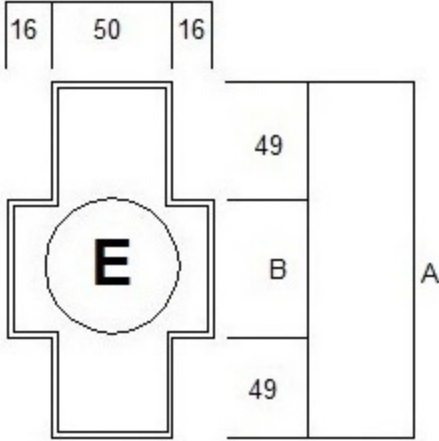
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	B	GIRTH FRONT	BRAKE LINE
12	124	26		
13	140	42		
14	145	47		
15	150	52		
6	156	58		
7	184	86		
8	190	92		
10	240	142		
15	265	167		
16	330	232		



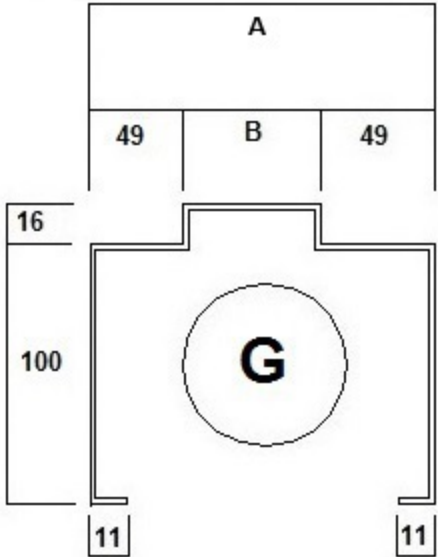
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	B	GIRTH FRONT	BRAKE LINE
14T	145	47		
7T	184	86		
8T	190	92		
18T	250	152		



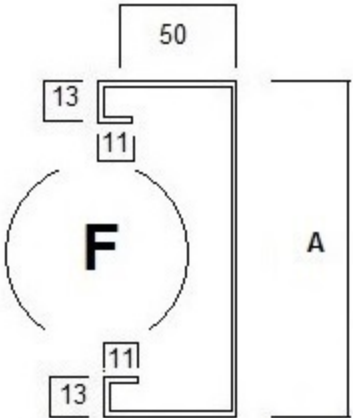
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	B	GIRTH FRONT	BRAKE LINE
15E	150	52		
6E	156	58		
8E	190	92		



"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

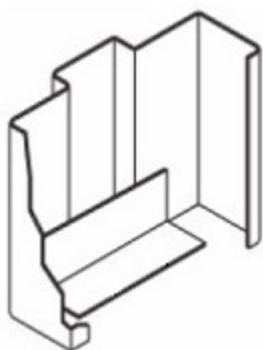
SECTION	A	B	GIRTH FRONT	BRAKE LINE
15G	150	52		
6G	156	58		
8G	190	92		



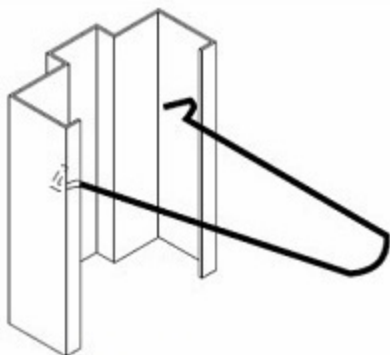
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	GIRTH FRONT	BRAKE LINE
12F	124		
13F	140		

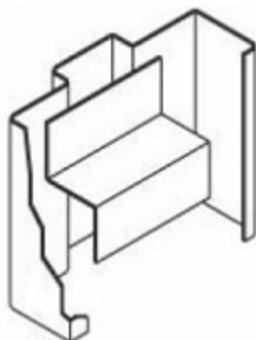
RECOMMENDED ANCHORAGE



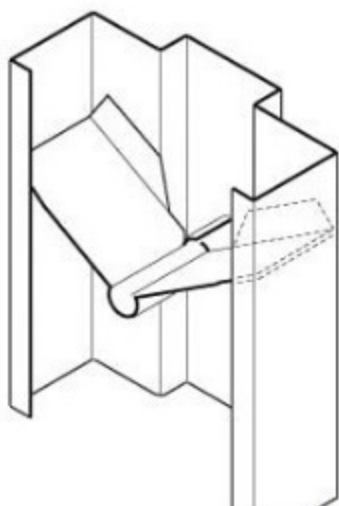
WELDED FLOOR ANCHOR (T/A)



MASONRY WIRE ANCHORS SHIPS LOOSE
WITH WELDED (T/A)



DRYWALL CHANNEL STUD ANCHOR (C/S)
WITH WELDED (T/A)



SCREW FIXED - EMA (S/F)
WITH GUIDE AND FRAME SUPPORT WELDED IN
3/8 FASTENERS SUPPLIED BY OTHERS



All Anchorage to be confirmed on shop drawings.
6 anchors per frame or screen up to 2286mm
8 anchors per frame or screen above 2286mm

Some anchors are shipped loose.

Extra or lost anchors can be ordered extra to contract.

 MATHESON CONSTRUCTORS	
Reviewed	(✓)
Reviewed as Noted	()
Revise and Resubmit	()
Not Reviewed	()
Date: 2026-07-07	
Reviewed by: 	
This review by MATHESON CONSTRUCTORS LIMITED is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of engineering elements which form part of the contract documents prepared by others. This review shall not mean that MATHESON CONSTRUCTORS LIMITED approves the detailed design inherent in the shop drawings, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the shop drawings or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.	

STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

GUIDE SPECIFICATION FOR INSTALLATION AND STORAGE OF HOLLOW METAL DOORS AND FRAMES



Canadian Steel Door
Manufacturers Association

L'Association Canadienne
Des Fabricants de Portes D'Acier

18 King Street East, Suite 1400, Toronto ON Canada M5C 1C4
Tel: (416) 363-7845 · Fax: (416) 369-0515 Email: marshl@marsh-executive.com
Website: www.csdma.org

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STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

PART I - INTRODUCTION

This guide specification is intended to stress the necessary precautions and requirements for receipt, storage, handling and installation of hollow metal products, and the skills essential in the hanging of doors.

The proper performance of most products depends not only on how they are manufactured, but how they are installed. This is especially true of hollow metal doors and frames. The installation of doors and frames is an operation demanding care and skill, if the doors are to operate properly. Care in manufacturing does not, in itself, guarantee satisfactory performance. Even the best designed and most carefully constructed frames and doors, if incorrectly installed, will not function properly.

Hollow metal work is fabricated in accordance with the shop drawings, approved by the architect or engineer. Preparation for hardware or other items supplied by others is provided in accordance with the information furnished to the hollow metal manufacturer.

The hollow metal manufacturer is a material supplier, not a subcontractor. The manufacturer does not include the installation of their product in the building, but only shipment in good condition from the factory.

Should the General Contractor discover any error in the hollow metal delivered to the job site, it is imperative that the hollow metal manufacturer be notified in writing and allowed sufficient time before initiating any corrective measure in the field, so that the manufacturer can participate in solving the problem. Failure to do so could result in the cancellation of the warranty and/or fire label and non-acceptance of any cost associated with repair.

Most member companies of the Hollow Metal Manufacturer's Association Division of CSDMA have their own field representatives who are qualified not only to do expert repair work but to determine whether the fault lies with the manufacturer or with some other party.

It is essential that material is properly stored prior to installation and skills are exercised in the setting of frames and hanging of doors.

PART 2 - RECEIVING AND STORAGE OF MATERIAL

A. RECEIVING

Upon delivery, the contractor responsible for receiving hollow metal products shall thoroughly inspect for damage. Cardboard and other wrappings shall be removed for inspection and to promote air circulation. Any scratches or disfigurements caused in shipping or handling shall be promptly cleaned and touched up with a direct to metal rust inhibitive primer.

Should damaged material be found, the General Contractor has the option of refusing delivery or to accept the material as damaged. For coordination purposes, CSDMA suggests that delivery should not be refused, but rather accepted as damaged. Any damaged items should be noted on the freight bill. Claims will not be honored by the freight carrier, unless the damaged items are noted on the freight bill at the time of delivery. The contractor shall notify the hollow metal manufacturer in writing immediately of any item signed for as damaged.

B. ON SITE STORAGE

Proper storage of hollow metal work at the construction site will help prevent damage to the primer coat of paint. Prime coated steel must be protected when exposed to the elements such as high humidity, salt air, rain, snow, and/or damp wrappings etc.... Particular attention must, therefore, be given to steel products having a coat of factory applied primer. Primer is porous to properly receive and hold top coats. Water or moisture, in contact with primer coated steel will seep through to the steel. An electrolytic action then follows, resulting in corrosion and causing the paint film to lose adhesion. The presence of oxygen at the water-air interface behind the loosened paint film accelerates corrosive action and the prime coat further deteriorates.

Even when zinc coated steel is used to provide corrosion resistance, manufacturers of hollow metal door and frame products have found that one week of product exposure to water, due to improper storage, can be equivalent to at least a year of outdoor exposure to the elements.

NOTE: Paint manufacturers advise that the primer typically used by hollow metal manufacturers should receive a finish coat of paint within 30 days of delivery. It is the responsibility of the General Contractor to sand, touch up and clean prime painted surfaces prior to finish painting in accordance with the finish paint manufacturer's instructions.

The following procedures shall always be observed in storing hollow metal doors and frames at the job site:

1. Store all materials in a dry area, under cover. All products shall be stored where they will not be exposed to, or come in contact with the elements.
2. Do not use non-vented plastic or canvas. These materials create a humidity chamber, which promotes blistering and corrosion.
3. Store doors and frames in an upright position with heads uppermost, Figures 1 and 2.
4. Place no more than 5 doors or welded frames in a group. Small groups not only minimize the likelihood of damage due to excess handling, but also facilitate selection from the group for installation. In the case of multi-opening frames, no more than three units should be stored in a group, to avoid serious damage to the bottom most frame.
5. Place all material on planking or blocking at least 4 in. (100 mm) off the ground, 2 in. (50 mm) off a paved area or the floor slab.
6. Provide at least 1/4 in. (6.4 mm) space between all units to permit air circulation.

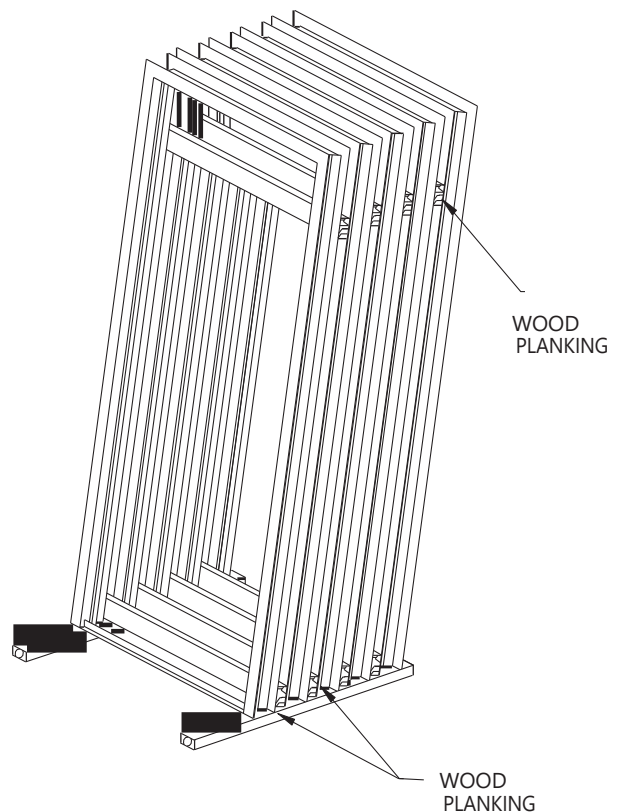


FIGURE 1
FRAME STORAGE

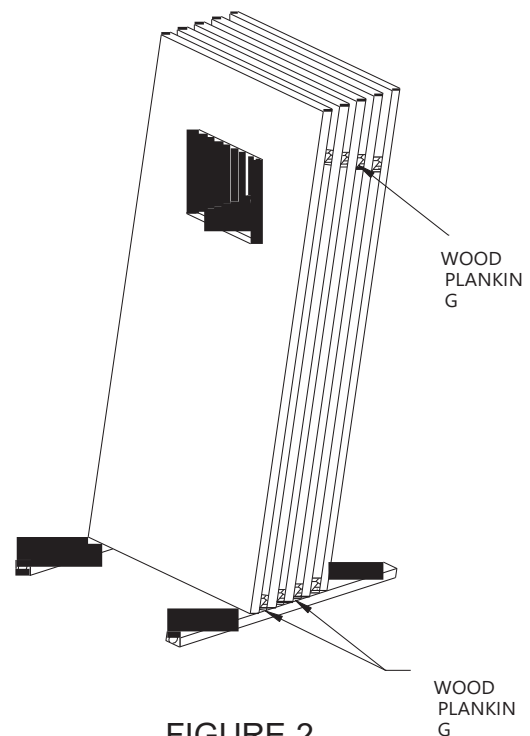


FIGURE 2
DOOR STORAGE

PART 3 - INSTALLATION OF FRAMES

A. GENERAL

Welded door frames are checked at the factory to ensure that they are square and that no jamb twists have occurred during fabrication. Temporary steel spreaders are then attached to the jamb base to minimize misalignment or other damage during handling and shipment. The frames are loaded on the carrier by personnel experienced in frame packing. The spreaders are for shipping and handling purposes only and must be removed before installing the frame.

At no time shall the spreaders be used during installation.

In spite of precautions, frames can and sometimes do arrive at the job site with minor alignment deviations. Minor deviations from true form and alignment can be corrected by the contractor responsible for installation.

When knock-down frames are used the manufacturer's recommended assembly procedures must be followed before installation of the frame can begin. A knock-down frame is easier to pack for shipment and requires less shipping space than a welded frame. Such frames, however, place greater responsibility on the installer, who must assemble them and make sure that, in spite of job conditions, they are square and in true alignment. Frames of this type normally use bendable tabs, screw or bolt fasteners in their assembly, and these may be lost in shipment or at the job site. In general, knockdown frames are not recommended for doors wider than 3 ft. 6 in. (1067 mm) or weighing more than 100 pounds (45 kg).

B. PRIOR TO INSTALLATION

The installer shall perform the following prior to installation: The

area of floor on which the frame is to be installed and the path of the door swing shall be checked for flatness and levelness. Permissible tolerance is +/- 1/16 in. (1.5mm). If the floor exceeds this, it is the general contractor's responsibility to correct the area that is out of tolerance before the frame is installed.

Frame shall be checked for correct size, swing, fire rating and opening number. Remove temporary steel spreaders. Spreader can typically be removed with cold chisel and hammer. With frame standing on concrete, position cold chisel at weld joint of spreader and jamb and strike chisel with hammer.

C. INSTALLATION TOLERANCES

During the setting of the frame, check and correct as necessary for opening width, opening height, squareness, alignment, twist and plumbness. Permissible frame product installation tolerances shall be maintained within the following limits: See Figure 3.

Opening width – Measured horizontally from rabbet to rabbet at top, middle and bottom of frame; +/- 1/32 in. (1mm).

Opening height – Measured vertically between the frame head rabbet and top of floor or bottom of frame minus jamb extension at each jamb and across the head; +/- 1/16 in. (1.5mm).

Squareness – Measured at rabbet on a line from jamb, perpendicular to frame head; not to exceed 1/16 in. (1.5mm).

Alignment – Measured at jambs on a horizontal line parallel to the plane of the face; not to exceed 1/16 in. (1.5mm).

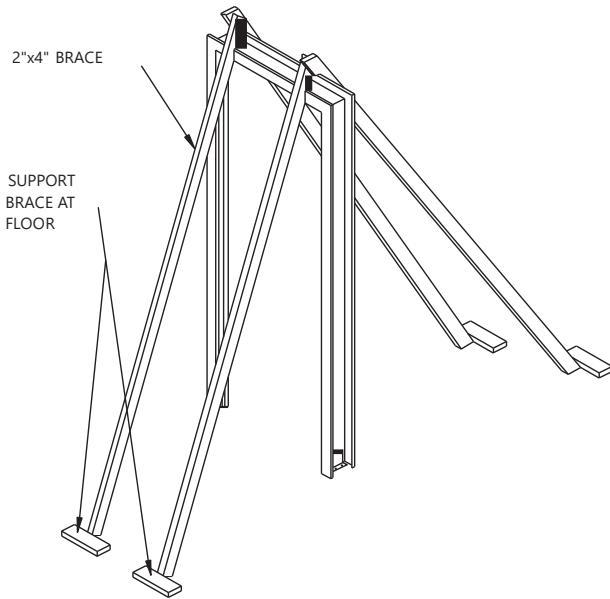
Twist – Measured at opposite face. Corners of jamb on parallel lines, perpendicular to the plane of the door rabbet.

Plumbness – Measured at jambs on a perpendicular line from the head to the floor; not to exceed 1/16 in. (1.5mm).

The tolerances shown provide a reasonable guideline for proper installation of hollow metal frame products. However, it should be noted that the cumulative effect of the installation tolerances at or near their maximum levels could result in sufficient misalignment to prevent the door from functioning properly. Installers should be careful not to create a tolerance buildup. Tolerance buildup occurs when several tolerances are at or near their maximums. Care should be taken to keep each of these tolerances as close to zero as possible.

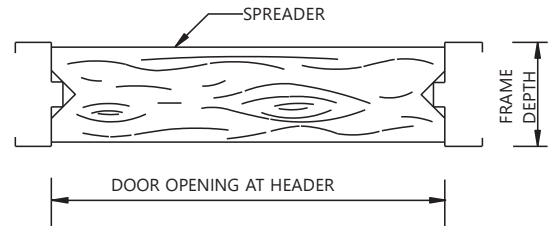
D. TYPICAL INSTALLATION PROCEDURES

Position frame in the correct location. Brace the frame as shown, Figure 4. Do not brace in the direction of intended wall.

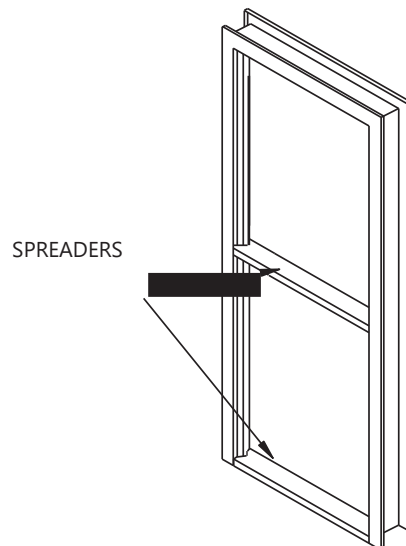


**FIGURE 4
FRAME BRACING**

With frame in position, install the temporary wood spreaders. The wood spreader, Figure 5, must be square and no less than 1 in. (25 mm) thick. Correct length is the door opening width between the jambs at the header. Cut clearance notches for frame stops. Spreader must be nearly as wide as frame depth for proper installation. Install a spreader at the bottom of the frame and a second wood spreader at the mid or strike point to maintain a proper door opening and to prevent bowing of the jambs, Figure 6. Clamp or wire spreaders to frame to hold spreaders in place until the frames are set permanently in the wall.



**FIGURE 5
WOOD SPREADER**



**FIGURE 6
SPREADER LOCATION**

At frames with jamb opening heights greater than 8 ft. (2438mm) or frame face dimensions less than 1-1/2 in. (38mm), install an additional wood spreader. Space wood spreaders at a maximum of 36 in. (914mm) intervals between header and bottom of frame, Figure 7.

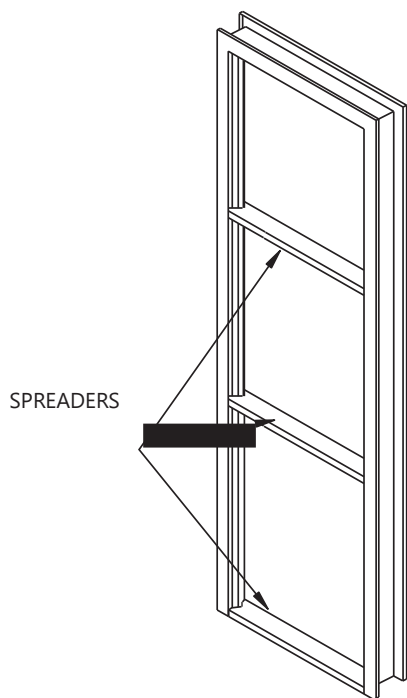


FIGURE 7
SPREADER LOCATION AT LARGE
JAMB OPENING HEIGHTS

At frames with sidelights where the sidelight sill intersects the door jamb near the strike, it is imperative that a wood spreader is located at this location, Figure 8.

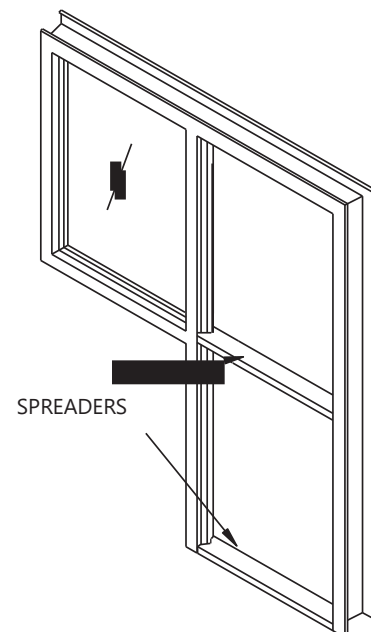
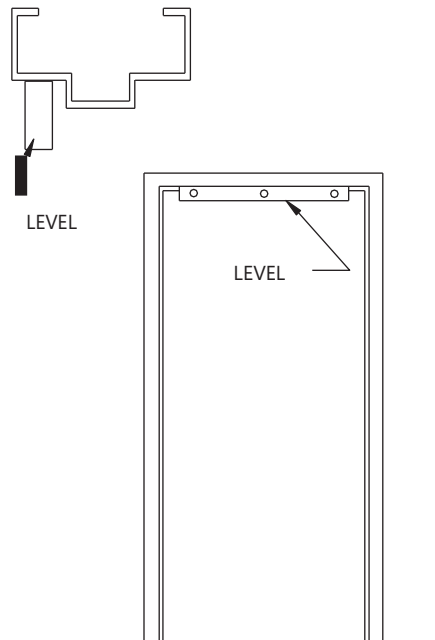


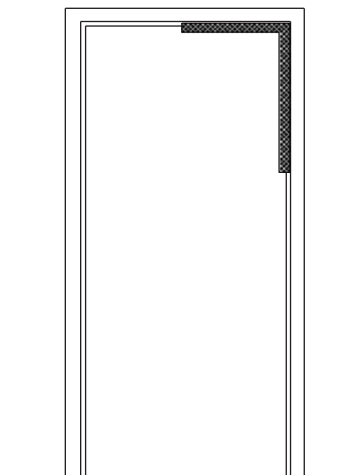
FIGURE 8
SPREADER LOCATION AT SIDELIGHTS

The installation contractor shall have a carpenter level and builder's square. Level the head by positioning the level to the head door rabbet, Figure 9. If necessary, adjust for high spots in floor by shimming under the jamb floor anchor, Figure 13a. Equalize them through an adjustable floor anchor, if specified, Figure 13b. Note, for labeled openings the maximum floor clearance is 3/4 in. (19mm).



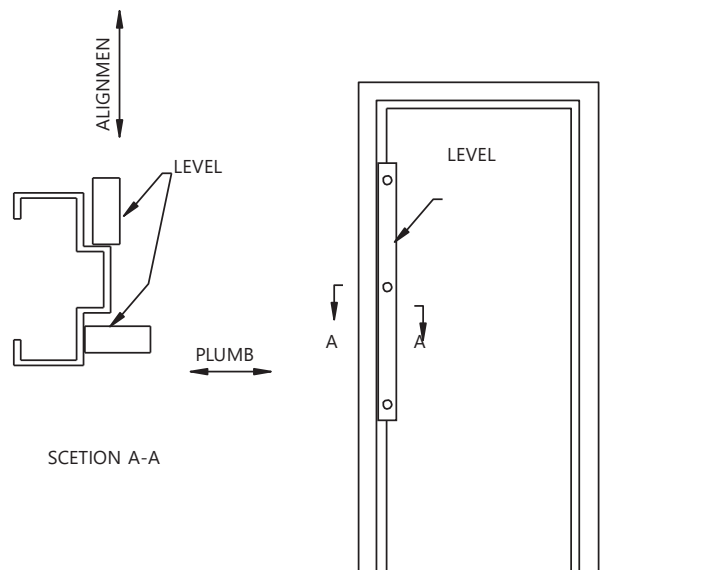
**FIGURE 9
LEVELING THE HEAD**

With builder's square, check frame for squareness. Position square against jamb and head at door rabbet, adjust as required, Figure 10.



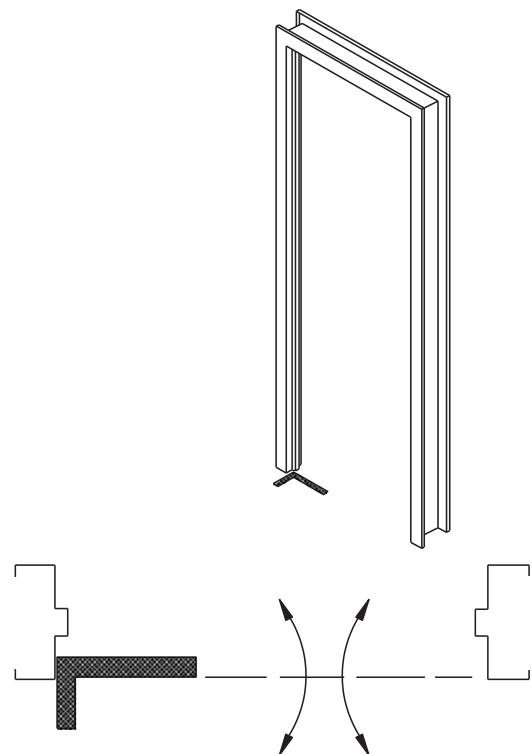
**FIGURE 10
SQUARENESS**

With carpenter level check the frame for plumbness and alignment: For plumbness, position level against both hinge and strike jambs in the rabbet. For alignment; Position level against both hinge and strike jambs on the stop, adjust as required, Figure 11.



**FIGURE 11
PLUMBING AND ALIGNMENT**

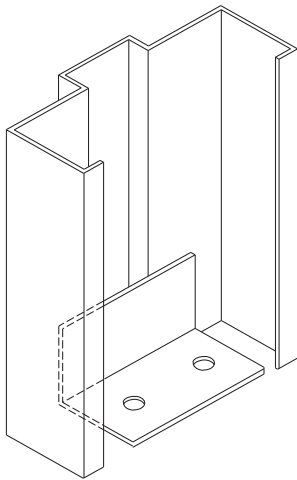
With builder's square, check jambs for twist. Position square against door rabbet and project line perpendicular to the plane of the door rabbet, adjust as required, Figure 12.



Once the installer has ensured that the frame is in the correct position, anchor the jamb to the floor.

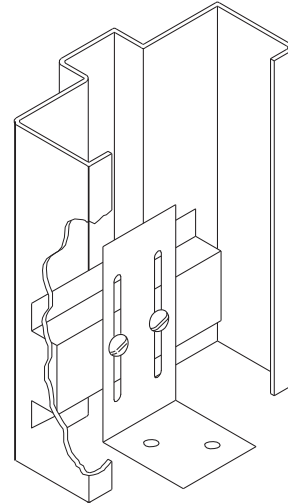
Floor Anchors: The fixed floor anchor is welded to the base of the jamb, typically secured to the floor by mechanical fasteners, providing solid anchorage for the base of each jamb, Figure 13a. Shimming must be used if the floor is not level. This type of anchor is not used in existing masonry, preframed stud walls or slip-on dry-wall frames, but may be provided.

At prepared openings (existing masonry or concrete) an additional expansion anchor reinforcement is provided. With pre-framed stud walls or completed drywall openings an additional base anchor is included. In both instances these anchors are located as close to the bottom of the jamb as is practical.



**FIGURE 13A
FIXED FLOOR ANCHOR**

The adjustable floor anchor is designed for use where there are significant irregularities from level, an intended slope in the floor, or where toppings such as terrazzo are used, Figure 13b. Like the standard anchor, it cannot be used in prepared openings or slip-on knock-down dry-wall frames.



**FIGURE 13B
ADJUSTABLE FLOOR ANCHOR**

E. WALL CONSTRUCTIONS AND APPROPRIATE ANCHORS

Proper frame anchoring is vitally important to the proper performance of the door opening. A variety of jamb wall anchors are available to suit the various types of wall construction. Anchors shown on the following pages are but a few of the variety of anchoring methods available.

ANCHOR SPACING

Anchor Spacing: CSDMA specifications require that the number of wall anchors used on each jamb, in all cases, be as follows:

- A. Openings in Masonry Walls. Frames with expansion bolt anchors; Anchors spaced a maximum of 6 in. (152 mm) from the top and bottom, with intermediate spacing at a maximum of 26 in. (660 mm) o/c. See Table 1.
- B. Openings in stud partitions with steel or wood stud anchors: Near hinges and directly opposite on strike jamb. See Table 1.
- C. Anchor quantity for slip-on drywall frames shall be as per manufacturer's standards and label requirements.
- D. Wall anchors locations and quantity are in addition to floor anchors, when required.

Table 1

Frame Anchor Spacing

Frame Height	Masonry Walls	Stud Partitions
> 0 in. < 60 in. (1524 mm)	2	3
> 60 in. < 90 in. (2286 mm)	3	4
> 90 in. < 96 in. (2438 mm)	4	5
> 96 in.	4 + 1 per 24 in. (610 mm)	5 + 1 per 24 in.



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18 King Street East, Suite 1400, Toronto ON Canada M5C 1C4
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Website: www.csdma.org